

RESEARCH ARTICLE

A Conceptual Model for Project Observatories

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ABSTRACT The use of observatories in various fields of knowledge has added a relevant characteristic to them, typological diversity. This has created the need to develop models that assist in the conceptualization and design of these observatories, considering the specificities of their application contexts. Thus, the main objective of this work is to propose a conceptual model for project observatories, considering aspects related to their structures, processes, and involved actors. To achieve the proposed objective, this research was conducted in two phases: 1) conception and 2) evolution and evaluation. The conception phase, which gave rise to the first version of the model, included an ad hoc literature review and a systematic mapping of the literature on observatories, as well as the development of ten pilot projects on project observatories. During the second phase, the model was evaluated and evolved into three additional versions through three focus groups, a survey, and a multiple case study. The final version of the model, named Model for Project Observatories (MPO), is organized from a set of 61 concepts structured into three hierarchical levels (general, intermediate, and specific concepts).

INDEX TERMS Conceptual model for observatories, project management, project observatories, observatories, transparency.

I. INTRODUCTION

The establishment of mechanisms to promote transparency has been a challenge for organizations [1]. The need to develop studies to investigate ways to strengthen the links between organizations and their external environments, improving services and promoting effective dialogue through technology, especially information systems, are highlighted in [2]. Current organizations face challenges regarding the transparency of their processes, as the transparency offered is insufficient, allowing only a small group to understand the business logic [3]. Transparency is presented by [4] as an emerging issue in information systems engineering.

In the context of projects, transparency has gained prominence in recent years [5]. The implementation of transparency in projects can vary depending on the type of project under development. For example, in public projects, such as governmental ones, the need for mechanisms that

promote transparency is evident and easily identifiable [6]. However, it is possible to see that this need for transparency is present to a greater or lesser extent in all types of projects [5].

In this scenario, observatories emerge as information systems capable of promoting transparency by supporting the collection, organization, storage, analysis, and disclosure of observations [7], [8]. The diversity of observatories in various fields of knowledge results in a wide range of models, aiming to aid the development of these initiatives [9].

Effective project observation, facilitated by tools such as project observatories, provides clear visibility into project activities and data. This visibility can lead to improved risk management, by allowing an enhanced understanding of risk propagation within project portfolios [10]. Furthermore, it can foster environments conducive to knowledge sharing and open-mindedness, crucial aspects for innovation in complex project contexts, as discussed in recent studies [11]. Project leadership, especially the soft skills of project managers like leadership, plays a vital role in encouraging knowledge sharing among team members and in optimizing

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knowledge management processes [12]. This increased visibility and the consequent sharing of information, essential for project success, are catalyzed by an organizational culture that values knowledge creation and capture [12].

Although there is a considerable presence of studies on observatories in the literature, including specific models presented by various authors such as [13], [14], [15], and [16], none of these models specifically addresses project observatories. Therefore, this research seeks to answer the following question: how can a conceptual model for project observatories be structured? The proposed solution is the formulation of a conceptual model, named MPO (Model for Projects Observatories), which contributes to the understanding and development of these observatories.

Within this context, the main objective of this work is to propose a conceptual model that contributes to the understanding and development of project observatories, addressing aspects related to their structures, processes, and involved actors. To achieve this objective, the following specific objectives were outlined: (1) Understand the development and characteristics of observatories; (2) Identify structural aspects of project observatories; (3) Identify processes that can be executed in (and by) project observatories; (4) Identify the actors involved in project observatories and their motivations; (5) Design a conceptual model for project observatories; (6) Evaluate and improve the proposed model.

II. BACKGROUND

A. TRANSPARENCY IN THE CONTEXT OF MANAGEMENT AND PROJECTS

With the advancement of digital technology, organizations have gained the ability to generate, process, and transmit large volumes of information at a marginal cost. These technological advancements, combined with other macroeconomic phenomena such as globalization and increased distrust in institutions, have led to unprecedented expectations for organizational transparency from the public [17]. According to [18], organizational transparency can be understood as the systematic disclosure of information that meets the needs of other actors.

In both public and private contexts, transparency establishes a set of aspects that suggest the existence of policies, procedures, and technologies to provide not only access, but also use, quality, understanding, and auditability of processes and information [19]. Generally, when discussing transparency in projects, it tends to be associated exclusively with public projects, with the disclosure of information to citizens [6], or with open-source software development projects, with the total openness of project artifacts to the entire community [20]. While transparency in projects is significant in these two situations, when examining the underlying concepts of transparency as presented by [21] and the categorization of transparency in pairs as reported by [22], it is evident that the need for transparency is present

in the daily life of all projects, not just public or open-source ones.

According to [5], the way transparency is implemented in projects is influenced by the dynamic evolution of project management theory and practice – from classical project management (mainly represented by the Project Management Body of Knowledge (PMBOK)) to the flexible, iterative-incremental methods of project and product management (Agile). The value of transparency and its operationalization in projects can vary across different existing management approaches.

In the classical project management approach, transparency is primarily reduced to the function of monitoring the project and informing all stakeholders using traditional forms of managerial communication (meetings, notices, formal project reports). Using the PMBOK [23] as an example of a classical approach, and based on [21] understanding of the underlying constructs of transparency (monitoring, process visibility, vigilance, and disclosure), it is possible to identify in this approach some knowledge areas aimed at implementing transparency in project management. These include Project Communications Management and Project Stakeholder Management. Furthermore, processes related to the Monitoring and Controlling Process Group also serve the purpose of implementing project transparency. According to [22] characterization, methods and tools within these knowledge areas and process groups reflect mechanisms of upward, downward, external, and internal transparency.

In traditional project management methodologies, transparency appears but is considered a second-order value; it is recommended but not mandatory. However, with the publication of the Agile Manifesto [24], transparency gained more prominence in project management, being considered a fundamental construct for most agile methods. In Scrum—as an example of an agile approach (or framework)—transparency is considered one of its pillars (along with inspection and adaptation) and signifies clarity of Scrum processes [25], meaning that significant aspects of the process must be visible to those responsible for the results.”

B. OBSERVATORIES

In the literature, there is no defined consensus on the concept of “observatories” [9]. However, it is possible to identify some common characteristics in studies related to the topic, which can assist in constructing a definition for these entities. One of these characteristics is understanding observatories as mechanisms that facilitate observation and, thus, promote transparency regarding a specific phenomenon, sector, or theme [26]. From this characteristic, other facets that contribute to the conceptualization of observatories can be derived.

As transparency tools, observatories play an important role by providing the community with access to data, information, and knowledge available about a specific phenomenon,

sector, or theme [27]. Additionally, they have a crucial role in knowledge management, encompassing everything from the creation to the sharing of knowledge, serving as support for decision-making for a given community [28]. Another characteristic of observatories is their ability to create a network or community, promoting interaction and dialogue among participants, thereby becoming instruments that stimulate collaboration among those involved [29].

The concept of observatories is closely related to the concept of information systems, which are composed of interrelated components that perform functions of data collection (or retrieval) [30], processing (or analysis) of this data [31], storage (or consolidation) of the information, and sharing (or dissemination) of the results [32].

After a comprehensive analysis of studies published in the literature, we propose the following definition for observatories: they are instruments, mechanisms, processes, or organizational units that enable the observation of a specific object, promoting transparency through the collection, consolidation, storage, study, research, analysis, sharing, monitoring, and dissemination of relevant data, information, and knowledge for a specific community.

C. MODEL DEFINITION AND CONCEPTUAL MODELS

To comprehend conceptual models, it is essential to first understand the concept of a model itself. Reference [33] assert that a model can be understood as a simplified representation through which certain aspects of reality are envisioned and better understood. Corroborating this, [34] define models as communication instruments where information is acquired and transferred through a representational structure.

[35] states that a model serves many purposes, but fundamentally it communicates something about the object of modeling to generate a more complete understanding of reality. This same author further presents the main functions of models as: i) an acquisitive function, through which information of interest is collected and ordered; ii) a logical function, which explains how a phenomenon occurs; iii) a normative function, which allows comparing phenomena to similar ones; iv) a systematic function, through which reality is described as a set of interconnected systems; and v) a constructive function, through which theories and laws are built.

In this sense, a conceptual model can be understood as a high-level description of how a system is organized and operates [36]. A conceptual model is frequently used to demonstrate a system's general functionality, interpreting the real world of the system with its specifications and with a representation, which can be achieved through textual description, but preferably through graphical means [37].

III. RESEARCH METHODS

The methodological design of this research was inspired by the research strategies defined in [38] and [39], combining

primary and secondary studies as a mechanism to obtain scientific evidence. The defined methodological design includes ten steps, grouped into two phases, as presented in Figure 1.

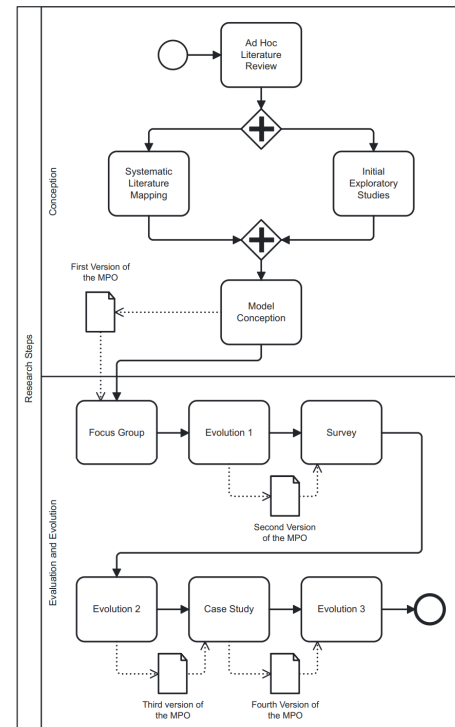


FIGURE 1. Research steps.

The first phase (Conception) aims to conceive the project observatory model. For this, a first version of the model was developed. The second phase (evaluation and evolution) incrementally seeks to evolve the proposed model in phase 1 into new versions. The objectives of each stage are presented below:

- **Ad Hoc Literature Review:** this stage sought to understand the concepts related to observatories and projects, contributing to the identification of the gap in the literature on the application of observatories in the context of projects.
- **Initial Exploratory Studies:** in this stage, the results of the ad hoc review were used as inputs for the execution of a set of exploratory empirical studies, which included the development of ten pilot project observatories.
- **Systematic Literature Mapping (SLM):** in this stage, the results of the ad hoc review were used as inputs for the execution of an SLM, which aimed to understand the state-of-the-art of observatories, seeking to understand how they have been developed and characterized in the literature. The results of this stage contributed both to substantiating the research gap and to confirming the facets of project observatories identified in the Initial Exploratory Studies.

- **Model Conception:** in this stage, the results of the Initial Exploratory Studies and the Systematic Literature Mapping were used as inputs for the construction of a first version of the MPO.
- **Focus Group:** in this stage, the first evaluation of the MPO (developed in the Model Conception stage) was carried out through three focus group sessions conducted with experts in observatories and projects.
- **Survey:** in this stage, the second evaluation of the MPO was conducted through a survey with participants from the Initial Exploratory Studies and authors identified in the Systematic Literature Mapping (SLM) on observatories.
- **Case Study:** the third and final evaluation of the model was conducted through the execution of a multiple case study, where the MPO was used in the development process of two project observatories.
- **Evolution 1, 2, and 3:** these stages aimed to evolve the model into new versions based on the evaluations carried out in the previous stages (Focus Group, Survey, and Case Study). These stages were responsible for evolving the first version of the model into intermediate versions (which were used as input in the Survey and Case Study stages) until reaching the final version of the model, produced after the Case Study stage.

IV. MODEL CONCEPTION

A. SYSTEMATIC LITERATURE MAPPING

1) PLANNING

The objective of this SLM is to investigate and understand the state of the art of observatories, aiming to identify their areas of application, definitions, characteristics, objectives, benefits, challenges faced, components, and, finally, the products and services offered by these observatories. This objective was broken down into the following main research question: “RQ. How are observatories being developed and characterized in the literature?”.

In an attempt to answer the main research question of the SLM, the following secondary questions were defined: RQ1. What are the application areas of observatories? RQ2. What are the definitions, characteristics, and objectives of observatories? RQ3. How can observatories support or benefit their areas of application? RQ4. What are the challenges faced by observatories? RQ5. What are the components of observatories? RQ6. What products and services are offered by observatories?

The SLM was planned based on the recommendations of [40]. As a search method, manual searching based on snowballing was chosen, following guidelines available in [41]. For data analysis, the thematic analysis guide presented in [42] was followed. Additionally, one inclusion criterion and ten exclusion criteria were defined to be used in the study selection stage.

Inclusion Criteria:

- **IC01** – Primary and complete studies that address at least one research question.

Exclusion Criteria:

- **EC01** – Editorials, tutorials, speeches, white papers, theses, dissertations, technical reports, and books.
- **EC02** – Articles that express personal viewpoints or opinions.
- **EC03** – Documents that are not complete articles or studies, such as presentations, web postings, web content, quotes, pamphlets, brochures, prospectuses, newsletters, or extended abstracts.
- **EC04** – Articles that do not address observatories.
- **EC05** – Articles that address observatories but are clearly unrelated to the research questions, thus falling outside the scope of this research.
- **EC06** – Secondary and tertiary studies.
- **EC07** – Articles that address astronomical observatories.
- **EC08** – Works that are not written in English, Portuguese, or Spanish.
- **EC09** – Works that are not available on the web for access and download.

2) EXECUTION

For this stage, three activities were performed: (1) construction of the initial list of works, (2) backward snowballing, and (3) forward snowballing. In stage 1, the recommendation presented in [41] was followed, and an exploratory search was conducted on Google Scholar to identify articles that addressed the Observatory theme.

From this initial search, 173 articles were identified. Next, duplicate articles were removed (first filter), resulting in a new list of 163 articles. After applying this first filter, all these articles were read, and the inclusion and exclusion criteria were applied. After applying the criteria, the initial set consisted of 65 articles.

One iteration of backward and forward snowballing was performed, the total of references and citations collected (2,329), after removing duplicate articles and applying the inclusion and exclusion criteria, 54 articles remained (22 from backward and 32 from forward). The final list of included works consists of the result of the backward and forward snowballing activities (54 articles) plus the 65 articles that composed the initial list of works, totaling 119 studies that advanced to the extraction stage. A list of all these included studies is presented in Appendix A.

3) RESULTS

RQ1. What are the application areas of observatories? Answers to this research question were identified in 118 (99%) of the 119 analyzed papers. The studies identified in this research presented observatories applied in the following areas: health (21/118), social (14/118), environment (13/118), web (12/118), media (11/118), science and technology (9/118), cities (8/118), social media (6/118), education and career (6/118), tourism (5/118), culture (3/118), projects (3/118), organization (3/118), industry (2/118), language (1/118), and religion (1/118).

RQ2. What are the definitions, characteristics, and objectives of observatories? The answers identified for RQ2 were grouped into the following themes: data, information, and knowledge (64/102), analysis (53/102), monitoring and control (48/102), environment (36/102), storage (30/102), communication (27/102), community (26/102), study, training, and capacity building (25/102), research (25/102), identification and collection (24/102), sharing (20/102), and observation (19/102).

RQ3. How can observatories support or benefit their areas of application? Answers to this research question were identified in 54 (45%) of the 119 analyzed papers. The evidence found in the literature in response to RQ3 was grouped into the following themes: enabling access to knowledge (30/54); supporting communication, cooperation, and collaboration (22/54), supporting planning and management (18/54), supporting businesses and organizations (18/54), supporting the development of studies and research (11/54), supporting society (11/54), and enabling innovation and future vision construction (5/54).

RQ4. What challenges do observatories face? Answers to this research question were identified in 25 (21%) of the 119 analyzed papers. The evidence identified in the studies responding to RQ4 was grouped into the following themes: challenges related to planning and management (9/25), challenges related to data collection and storage (8/25), challenges related to interoperability (7/25), challenges related to processing (6/25), challenges related to relationships (6/25), challenges related to maintenance (5/25), challenges related to users (5/25), and challenges related to security and privacy (4/25).

RQ5. What are the components of observatories? Answers to this research question were identified in 20 (17%) of the 119 analyzed papers. The answers identified for RQ5 were grouped into the following themes: products and services related to monitoring and analysis (15/20), products and services related to data repositories (8/20), products and services related to dissemination and sharing (7/20), and products and services related to the community (5/20).

RQ6. What products and services are offered by observatories? Only 5 studies (4% of the total analyzed studies) responded to this question. The answers identified for RQ6 were grouped into the following themes: components related to storage (5/5), components related to observation and collection (4/5), components related to visualization, analysis, and publication (2/5), and components related to management and quality (2/5).

4) CONTRIBUTIONS TO THE MODEL

The results obtained from the SLM were used to confirm the facets of project observatories identified from the execution of the initial exploratory studies (described in Section IV-B), giving rise to the first version of the MPO. Appendix B presents a traceability of the model elements to the themes identified in the SLM. More details on the SLM results are discussed in [43].

B. INITIAL EXPLORATORY STUDIES

1) PLANNING

The initial exploratory studies aimed to assist in understanding the functioning of project observatories through the execution of a set of pilot projects. To execute these studies, the availability of teams for the development of pilot projects was necessary.

Additionally, the exploratory and innovative nature of these projects required the participation of individuals with diverse experiences. For these reasons, the execution of these studies was planned in disciplines of a graduate program with the participation of master's and doctoral students.

Data collection took place during the studies, through observation conducted by the authors of this study and, at the end of the studies, from the documents produced by the teams and from the websites of the developed project observatories. The analysis of the collected data in the studies was conducted using the method of faceted taxonomic analysis, for which the simplified process for faceted analysis presented in [44] was employed.

2) EXECUTION

The studies were conducted across three distinct disciplines, each involving the development of a set of pilot project observatories. The first discipline, focused on Collaborative Systems, was where Study 1 (Universal Project Observatory) was developed. In this study, students were challenged to propose a collaborative environment supporting universal project information sharing, leading to the conception of an initial reference document for the Universal Project Observatory (UPO). Studies 2, 3, 4, 5, 6, and 7 were conducted in a second discipline, focusing on Project Management. All students in this discipline were technical-administrative staff from the Federal Institutes of Education, Science, and Technology (IFs) and UFPE. Due to this connection, participants in this discipline, divided into six groups, were tasked with developing pilot project observatories for their respective institutions. Finally, Studies 8, 9, and 10 were developed in a third discipline, which was exclusively dedicated to the development of project observatories and was its main theme. In this discipline, divided into three groups, participants developed three pilot project observatories.

3) RESULTS

Study 1: Universal Project Observatory. This study aimed to conceive an initial reference document to enable the development of a Universal Project Observatory (UPO). The construction of the observatory itself was not within the scope of this study. The initial reference document was developed based on a set of investigations carried out by the team. At the end of this study, a definition for the UPO was proposed, along with the identification of its main characteristics and requirements, and the construction of a screen prototype.

Study 2: Projects Observatory of the Northern Federal Institutes. This study aimed to develop a pilot project of

a projects observatory for the following Federal Institutes (IFs) in the Northern Region of Brazil: Federal Institute of Acre (IFAC), Federal Institute of Amazonas (IFAM), Federal Institute of Rondônia (IFRO), and Federal Institute of Tocantins (IFTO).

Study 3: Projects Observatory of the Northeastern Federal Institutes I. This third study aimed to develop a pilot project of a projects observatory covering the following Federal Institutes (IFs) in the Northeastern Region: Federal Institute of Ceará (IFCE), Federal Institute of Rio Grande do Norte (IFRN), Federal Institute of Piauí (IFPI), and Federal Institute of Maranhão (IFMA).

Study 4: Projects Observatory of the Northeastern Federal Institutes II. The fourth study aimed to develop a pilot project of a projects observatory covering the following Federal Institutes (IFs) in the Northeastern Region: Federal Institute of Bahia (IFBA), Bahian Federal Institute (Bahian IF), Federal Institute of Paraíba (IFPB), and Federal Institute of Pernambuco (IFPE).

Study 5: Projects Observatory of the Central-Western and Southeastern Federal Institutes. This study developed a pilot project of a projects observatory for the following Federal Institutes (IFs) in the Central-Western and Southeastern Regions of Brazil: Federal Institute of Espírito Santo (IFES), Federal Institute of Triângulo Mineiro (IFTM), Federal Institute of Brasília (IFB), Federal Institute of Goiás (IFG), and Federal Institute of Mato Grosso (IFMT).

Study 6: Projects Observatory of the Santa Catarina Federal Institutes. This study aimed to develop a pilot project of a projects observatory for the Federal Institutes (IFs) in the State of Santa Catarina: Federal Institute of Santa Catarina (IFSC) and Catarinense Federal Institute (IFC).

Study 7: Projects Observatory of the Federal University of Pernambuco. In this study, a pilot project of a projects observatory for UFPE was developed. This observatory was proposed as a tool to gather and manage various information regarding the institution's projects, strengthening the task of disseminating this information and also serving to present an overview of entrepreneurial initiatives at UFPE.

Study 8: University Entrepreneurship Projects Observatory. The University Entrepreneurship Projects Observatory (UEPO) was proposed as a space for observation, analysis, and reflection of movements in entrepreneurship and entrepreneurial initiatives in university centers, stimulating collaboration among professors, students, researchers, and entrepreneurs to build content and knowledge on the subject, driven by inspiring cases.

Study 9: PMI Pernambuco Projects Observatory. The PMI Pernambuco Projects Observatory was proposed as a solution to provide visibility to projects and actions carried out by the Pernambuco Chapter of the Project Management Institute (PMI-PE) to society as a whole, as well as to generate interaction among its members and internal chapters of the institute.

Study 10: Pernambuco State Public Works Projects Observatory. In this study, a pilot project was developed for

the Pernambuco State Public Works Projects Observatory (ObrasPE) with information on projects from the last 15 years audited by the State Court of Auditors of Pernambuco (TCE-PE).

Faceted Analysis. Understanding observatories from a sociotechnical perspective [15], i.e., from a set of users and technologies in a social context, concepts related to project observatories were structured into three main facets: Structures, Processes, and Agents. Appendix C provides more details on the results of the faceted analysis conducted in this study.

4) CONTRIBUTIONS TO THE MODEL

The first version of the MPO was constructed based on the results obtained from the initial exploratory studies and confirmed through the results achieved with the systematic literature mapping on observatories.

The facets and sub-facets, identified from the initial exploratory studies and confirmed by the systematic literature mapping, were organized into three hierarchical levels (dimensions, sub-dimensions, and elements), giving rise to the first version of the MPO. More details about the results of the initial exploratory studies are discussed in [45] and [46].

5) FIRST VERSION OF THE MPO

The first version of the MPO (Figure 2) was structured into three dimensions: Structures, Processes, and Agents. The dimensions group a set of sub-dimensions, which, in turn, congregate a set of elements. More details of this first version, including a description of each element, are presented in Appendix D.

V. EVALUATION AND EVOLUTION OF THE MODEL

A. FOCUS GROUP

1) PLANNING

For this study, a set of activities were defined—based on the activities presented in [47] and [48]—as follows. The focus groups developed in this study aimed to conduct an initial evaluation, from the perspective of project and observatory specialists, of the proposed model for project observatories, thus characterizing them as exploratory focus groups.

During the activity of planning the focus group, a pre-group questionnaire was developed (available in Appendix E). The questionnaire was sent to the invited participants along with a guidance document of the first version of the MPO. The themes to be discussed during the focus groups were defined, based on the feedback about the model collected through this questionnaire.

Two focus group sessions were planned: one with project specialists and another with observatory specialists. In addition to these two sessions, a session was planned with the objective of identifying improvements in the planning and conduction of the other sessions. This pilot session was conducted with researchers from the research group in which the authors of this work are involved, who have experience

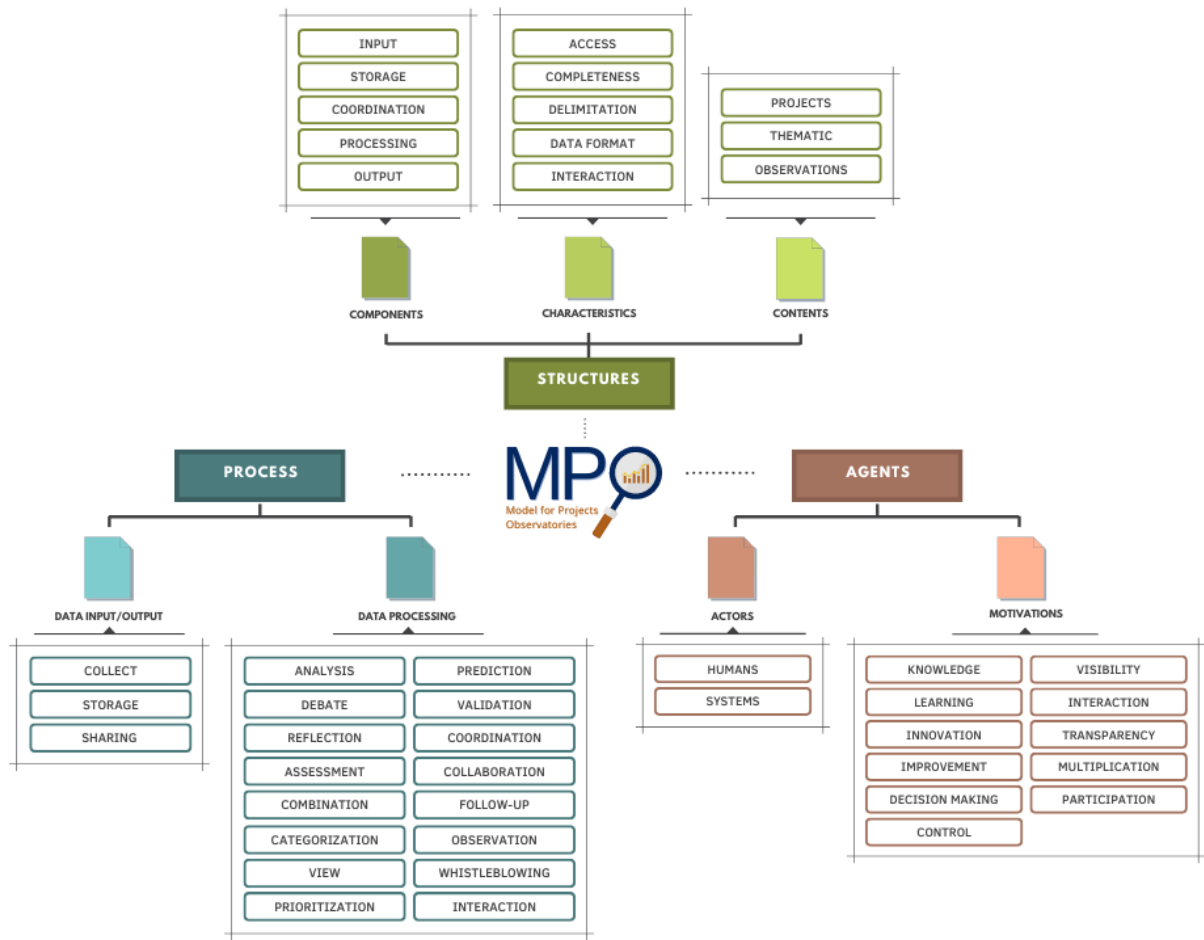


FIGURE 2. First version of the MPO.

in developing project observatories. After this pilot session, it was decided that the results collected from it would be used to compose the body of knowledge of this research.

Some desired characteristics for the participants were defined. For the group with project specialists, participants should have at least five years of professional and/or academic experience with projects. For the group with observatory specialists, participants should have at least two years of experience with observatories.

Data analysis was planned to be executed after conducting the focus groups, with the objective of identifying suggestions for improvements to the first version of the model. Following the guidance available in [47], the data were analyzed using the ethnographic summary technique.

2) EXECUTION

Twenty-one experts were invited to participate in three focus group sessions; however, only thirteen accepted the invitation. Of these, four participated in the first session (with research group participants experienced in project observatories), six participated in the second session (with project specialists), and three participated in the third session (with observatory specialists).

Through data analysis, it was possible to identify a set of improvements for the first version of the model, as suggested by the three focus groups. A draft of the second version of the model was constructed based on these improvement suggestions. This preliminary version was made available again for evaluation by the focus group participants through a questionnaire in the member-checking activity. This questionnaire was sent to all thirteen participants, and out of these, six responded.

3) RESULTS

Improvement Suggestions. Thirty-five improvement suggestions were identified from the analysis of the data collected in the focus groups. These suggestions were grouped into the following categories: model organization (6 suggestions); adjustments to nomenclature and descriptions (18 suggestions); addition of concepts (9 suggestions); and removal and merging of concepts (2 suggestions). In the first focus group, 18 improvement suggestions were identified; in the second group, 14 improvement suggestions emerged; and in the third group, 10 improvement suggestions were collected.

Model Evaluation. The focus group participants were asked if they believed that the first version of the MPO was

capable of aiding in the understanding and development of project observatories. Both the second and third focus groups responded positively to this question. However, the first focus group had partial agreement on this matter. The focus group participants were also asked to rate (on a scale of 1 to 5, where 1 corresponds to very poor and 5 represents excellent) the ease of understanding the model. In this regard, the overall average rating from the three groups was 3.8.

Member Checking. After making adjustments to the MPO based on the improvement suggestions collected from the focus groups, a draft of the second version of the model was made available to the focus group participants, this time in the form of an electronic questionnaire (available in Appendix F). The participants were asked if the suggestions made by their focus group were included in the new version of the model. Unanimously, all respondents confirmed that their improvement suggestions were fully addressed. Participants were also asked if they believed that the new version of the model would be capable of aiding in the understanding and development of project observatories. Once again, unanimously, all responded positively to this question. Finally, participants were asked to rate (on a scale of 1 to 5) the ease of understanding the new version of the model. In this regard, the overall average rating was 4.11, surpassing the average responses given by the focus groups for the same question in the evaluation of the first version of the model.

4) CONTRIBUTIONS TO THE MODEL

This activity generated a set of contributions to the model, among which the following can be highlighted:

- A new nomenclature was used for the organization of the model: General Concepts (Dimension), Intermediate Concepts (Subdimension), and Specific Concepts (Elements);
- More details were included regarding the model structuring process;
- Definitions were added for all concepts, including intermediate concepts;
- The entire model was reviewed to avoid the use of repeated nomenclature;
- A usage scenario was included for all specific concepts presented in the model;
- One new intermediate concept and fifteen new specific concepts were added;

Four specific concepts were removed. More details about the results of the focus groups conducted in this study are discussed in [49].

B. SECOND VERSION OF THE MPO

The second version of the MPO was developed based on the adjustments made to the first version according to the improvement suggestions collected from the focus groups. Figure 3 provides an overview of the second version of the model, with more details about this version presented in Appendix G.

C. SURVEY

1) PLANNING

A set of activities was planned to be executed in this survey. For this, the activities defined in [50] were used as a reference. This survey aimed to analyze the second version of the MPO with the purpose of evaluating the relevance of the concepts understood by the model from the perspective of experts in observatories and in projects. Based on this objective, the following research questions were delineated: RQ1: Is there any concept presented in the second version of the MPO that is not relevant in the context of project observatories? RQ2: Is it necessary to add any new concept or make any changes to the existing concepts in the second version of the MPO? RQ3: What is the order of relevance of the specific concepts included in the second version of the MPO? RQ4: Can the concepts included in the second version of the MPO contribute to the understanding and development of project observatories?

The questionnaire used in this survey for data collection was structured into six parts and can be found in Appendix H. This questionnaire was evaluated by two researchers who are experts in projects and observatories before being distributed to the respondents.

In this study, the experts are comprised of two groups. The first group (approximately 174 individuals) consists of the authors of the studies identified in the Systematic Literature Mapping on observatories (presented in Section IV-A). The second group (approximately 55 individuals) is formed by the participants in the pilot projects of project observatories developed in the initial exploratory studies (Section IV-B). Thus, the target population of this study comprises 229 individuals.

The data analysis was carried out in four stages: definition of weights for the participants, identification of improvements, prioritization of specific concepts, and overall evaluation of the model. Descriptive statistics were used as the data analysis method.

The definition of weights for the participants was inspired by the strategy adopted in [51]. This approach aimed to enhance the reliability of the survey results by attributing greater influence to the responses of more experienced participants. Specifically, participants with more experience were assigned a higher weight in the analysis of their responses. For this, three key criteria were used to construct this weight: academic degree, experience with observatories, and experience with projects.

The identification of improvements was based on the open-ended questions of the questionnaire. A set of criteria was defined, inspired in [51], taking into account the calculated weight for the participants, for the acceptance of the improvements suggested by the participants.

In the survey form, participants were invited to assess the relevance of each specific concept included in the model using a five-level Likert scale, ranging from level 1 (not relevant) to level 5 (extremely relevant). The relevance of each specific concept was then calculated as a weighted

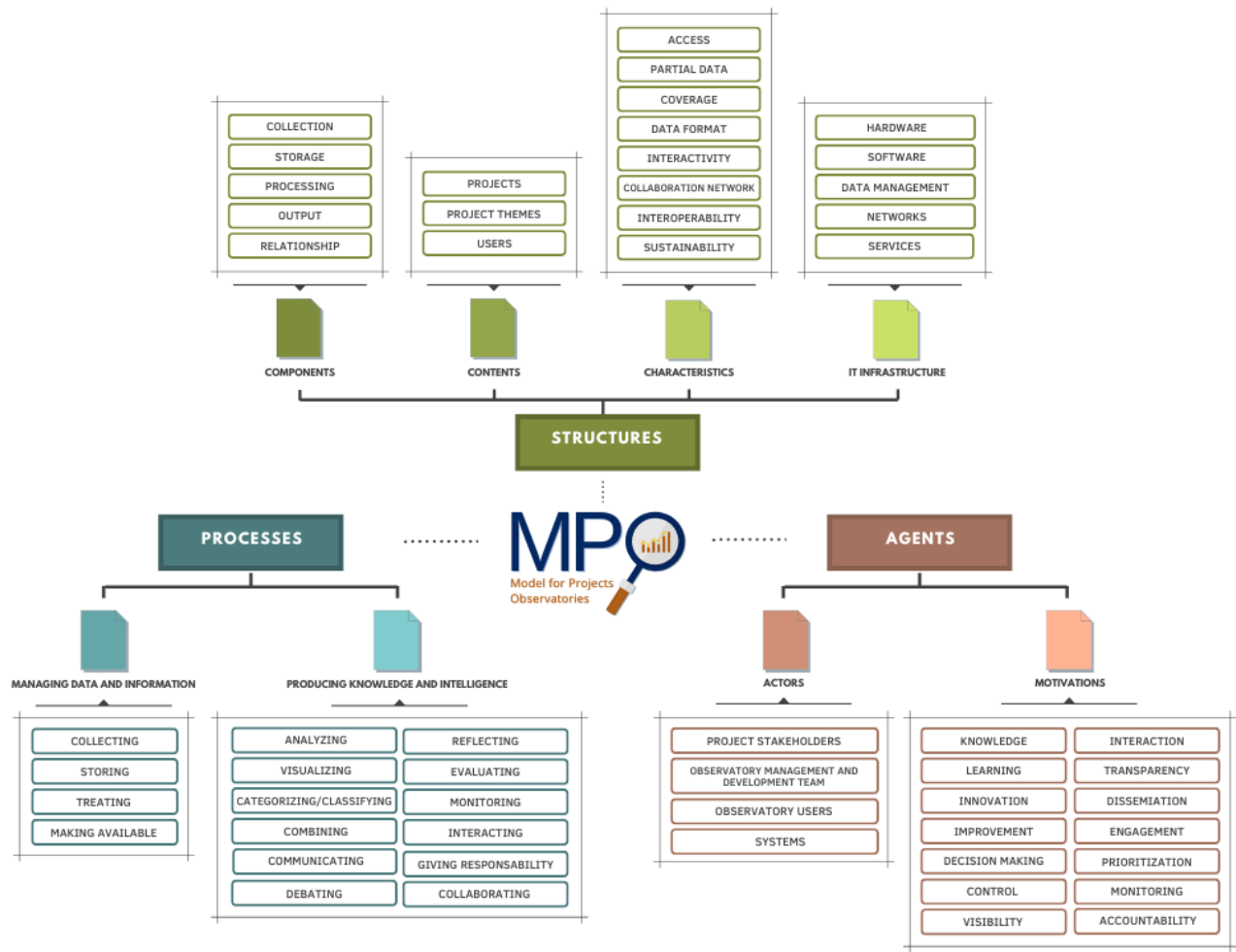


FIGURE 3. Second version of the MPO.

average of the ratings assigned by each participant, meticulously accounting for the individually determined weight of each participant. This rigorous weighted average method ensured that expert opinions with higher relevance (based on experience) had a proportional impact on the final concept prioritization.

2) EXECUTION

Out of the 229 invited to participate in the survey, 52 responded to the form, with 15 from the first group (authors of the studies identified in the SLM) and 37 from the second group (participants in the pilot projects of project observatories).

In total, 188 emails were sent without resulting in failures. This could be an indication that this would be the closest value to the real population of this study. Since 52 responses were collected, the confidence level of this study is around 88% and a response rate of 28%. Furthermore, to ensure the internal consistency and reliability of the survey instrument (questionnaire), it was rigorously submitted to Cronbach's alpha test. The obtained result of 0.96 indicated an almost

perfect internal consistency, thus validating the suitability of the instrument for collecting reliable data on concept relevance

3) RESULTS

Characterization of Participants: Regarding academic background, the majority of participants have computing as their field of study (71%), and most hold a master's degree (66%). Participants were asked about their experience with observatories. Almost all respondents (94%) stated they have academic and/or professional experience with observatories, with the majority (78%) having up to four years of experience. The vast majority of respondents (77%) stated they have academic and/or professional experience with projects, with most (58%) having up to ten years of experience.

Analysis of Concept Relevance: The specific concepts included in the second version of the MPO were evaluated by survey participants regarding their relevance. From the analysis of the responses, the aim was to identify the existence of any specific concept included in the second version

of the MPO considered not relevant by the participants. However, all concepts were considered very or extremely relevant by the majority of survey participants. Thus, it is believed that there is no need to exclude any specific concept.

Suggested Improvements: Participants were asked if they deemed it necessary to make any changes to the concepts included in the model. Applying the defined criteria for accepting suggestions resulted in a set of ten improvements that were implemented in the new version of the model.

Concept Prioritization: From the calculated relevance, it was possible to establish a relevance order for the specific concepts contained within the intermediate concepts. Thus, the specific concepts could be grouped into two relevance levels: Level 1 and Level 2. In Level 1, the specific concepts that had their relevance equal to or higher than the average relevance of all specific concepts of that intermediate concept are grouped, while in Level 2, the specific concepts with relevance lower than the average are grouped.

Model Evaluation: Survey participants evaluated the MPO from three perspectives: (1) its ability to assist in understanding and developing project observatories, (2) ease of understanding, and (3) practical applicability. Regarding the first evaluation perspective, almost all participants (96%) partially or fully agreed that the MPO can contribute to the understanding and development of project observatories. With respect to the second evaluation perspective, the vast majority of participants (92%) partially or fully agreed that the MPO is easy to understand. Finally, it was possible to observe, with the third evaluation perspective, that 90% of survey participants agreed (partially or fully) that the model is viable for practical application.

4) CONTRIBUTIONS TO THE MODEL

This activity generated a set of contributions to the model, among which it is possible to highlight:

- Specific concepts were grouped into levels of relevance;
- Merging of seven specific concepts resulting in three new concepts;
- Addition of two new specific concepts;
- Changes in the nomenclature and description of six specific concepts;
- Removal of four specific concepts.

Further details about the results of the survey conducted in this study are discussed in [52].

D. THIRD VERSION OF THE MPO

The third version of the MPO was developed based on the adjustments and enhancements made in the second version, using the results of the survey presented in Section V-C as a basis. Figure 4 provides an overview of the third version of the MPO, with further details of this version presented in Appendix I.

E. CASE STUDY

1) PLANNING

The following objective was delineated for this case study: to evaluate the third version of the MPO, from the description (thus characterizing itself as a descriptive case study), in real contexts, of how project observatories can incorporate the concepts presented in the model. The analytical technique used was cross-case synthesis [53].

The following research question was defined for this study: RQ. How can a project observatory be developed based on the concepts presented in the third version of the MPO? The steps taken in this study were based on the three-stage process for conducting multiple case studies defined in [54].

The following descriptive theoretical proposition was built for this study, following Yin's guidance [54]: the case study will describe how a project observatory can be developed based on the concepts contained in the MPO. A complete description of what should be studied should include (a) a description of how the MPO was used in the observatory development process; (b) a mapping of which model concepts were covered by the observatory; (c) a description of how the MPO concepts were covered by the observatory; (d) the identification of MPO concepts that were not covered by the observatory; (e) if necessary, the identification of new concepts for the MPO; and (f) the identification of difficulties in understanding the MPO.

This research included the study of two cases, with the unit of analysis defined as a project for the development of a project observatory. For this study, four data collection sources were defined: interviews with members of the observatories' development teams; participant observation conducted by the authors of this study during the development of the observatories; documentation (including reports produced by the teams, requirements documents, among others) produced during the observatories' development process; and analysis of physical artifacts, in this case, the developed project observatories and their source code.

2) EXECUTION

The choice of cases was based on them being common cases, meaning the cases present circumstances and conditions of an everyday situation [54]. Additionally, cases were selected for literal replication, that is, to predict similar results.

The development projects of the following project observatories were selected as cases to be studied in this research: Project Observatory of the Computer Practices Center (NPI) at the Federal University of Ceará (UFC) - Quixadá Campus (Case 1) and Project Observatory at the University of Pernambuco (UPE) (Case 2).

3) RESULTS

The following contextual information about the cases studied here is essential for understanding this synthesis: (1) the model was used, in both analyzed cases, by students in

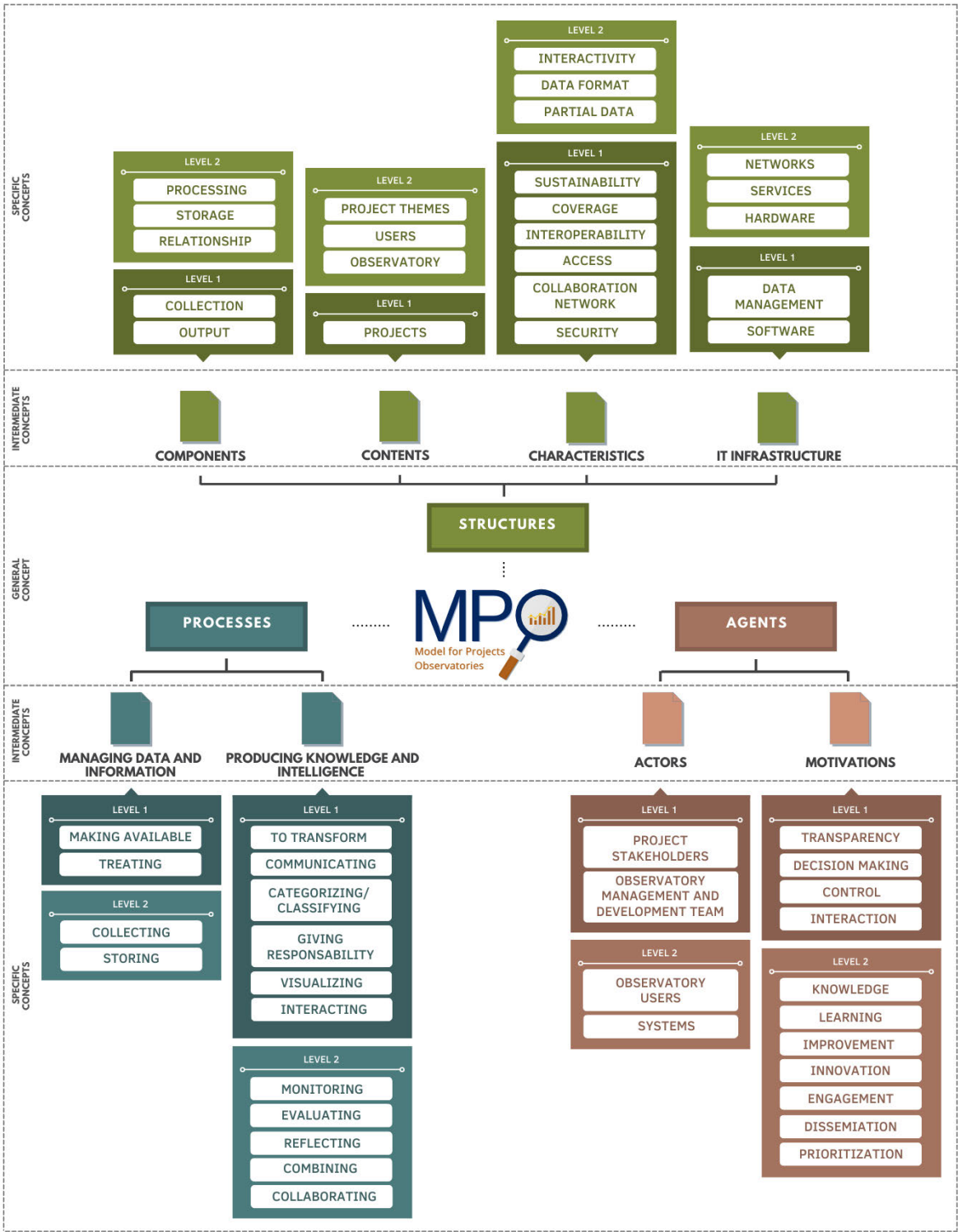


FIGURE 4. Third version of the MPO.

academic contexts; and (2) the observatories developed in the studied cases were scoped within public educational organizations (or sectors of these organizations). Thus, it is believed that other forms of using the model can be identified

from other cases with contexts different from those studied here.

The first research question of this study sought to identify how the MPO was used in the process of developing the

observatories. It was observed that, in both cases, the model was used throughout the entire development process of the observatory, with emphasis on the conception and evaluation stages. Since the NPI Project Observatory (Case 1) was developed in the context of a research project, the MPO was also used as a theoretical basis for the research. In both cases, activities of requirements collection, development activities, and evaluation activities were carried out. Appendix J provides more details about the responses identified in each of the cases for this research question.

The second research question of this study aimed to understand which and how the MPO concepts were covered by the observatories. The identified evidence indicated that 81% of the model's concepts were covered in Case 1 (NPI Project Observatory) and 91% in Case 2 (UPE Project Observatory). Appendices I and J detail how the two cases covered each of the specific concepts.

The third research question sought to understand which MPO concepts were not covered by the observatories. Comparing the two cases studied, it is possible to see that eleven concepts were not covered by at least one of the cases. Of this total, three ("Project Themes" from the intermediate concept "Contents"; "Networks" from the intermediate concept "IT Infrastructure"; and "Availability" from the intermediate concept "Data and Information Management") were not covered by either case. Appendix J provides more details on the concepts not covered by the cases studied.

The fourth research question sought to identify new concepts for the MPO from the cases studied. It was observed in both cases the need for a concept related to the ease of use of the project observatory.

Finally, the fifth research question sought to understand if there were difficulties, on the part of the observatory development team, in understanding the model. From the participant observation process conducted by the authors of this study during the execution of the cases, it was identified, for both cases, that the participants had difficulties in understanding the difference between the specific concepts of the intermediate concept Components and the specific concepts of the intermediate concepts Data and Information Management and Knowledge and Intelligence Production. During the conducted interviews, the participants of the studies were asked if they confirmed or not this perception, only the interviewee from Case 2 disagreed with this perception. Another difficulty identified in Case 2 was the development team's disagreement regarding the relevance classification presented in the third version of the model for some specific concepts.

4) CONTRIBUTIONS TO THE MODEL

The main contributions of this study to the new version of the MPO were: proposing a new classification of relevance and prioritization of the specific concepts presented in the model; adding the specific concept "Usability" to the intermediate concept "Characteristics"; including more information to clarify the differences and similarities between

the specific concepts contained in "Components" and the processes contained in the intermediate concepts "Data and Information Management" and "Knowledge and Intelligence Production."

During the analysis of the cases, it was observed that the boundaries between some specific concepts were not sufficiently clear. For this reason, it was decided that it would be more prudent to unify some specific concepts.

In summary, two case studies were conducted to evaluate the practical application of the MPO in real project observatories. The first case involved the development of the Project Observatory at the Nucleus of Informatics Practices (NPI) at UFC-Quixadá, carried out within an undergraduate capstone project. The MPO was used throughout the development process, particularly in the conception and evaluation phases, resulting in the incorporation of 81% of the specific concepts defined in the model. The second case study was conducted in partnership with the University of Pernambuco (UPE), involving graduate and undergraduate students. This observatory was designed to support transparency and integration across research, extension, and innovation projects within the institution, incorporating 91% of the model's specific concepts. In both cases, the MPO provided a comprehensive foundation for structuring, designing, and evaluating the observatories. These findings demonstrate the model's applicability and reproducibility in academic contexts. Additional implementation details are provided in Appendices I and J.

F. FOURTH VERSION OF THE MPO

1) PREMISES

The MPO represents a conceptual view rather than a normative definition. Therefore, it is not expected/required that a project observatory encompasses all the concepts presented in the model. Indeed, it can be expected that these concepts will evolve/adapt over time. Additionally, the MPO does not aim to provide a sequential implementation guide.

In an effort to prioritize the specific concepts contained in the model, these concepts were grouped into two categories: primary and secondary. The primary concepts are the most common to project observatories; they were included in all project observatories studied/developed within this research. However, given the model's flexibility and adaptability, these concepts are not considered mandatory, though they are highly recommended for inclusion in a project observatory. The secondary concepts are less common (compared to primary concepts) and were not identified in all the project observatories studied. However, this does not mean they should be disregarded or considered less relevant than the primary concepts (the intended observatory's context should be taken into account to define these issues). Instead, it may indicate that a project observatory can function without the need for these concepts. As another form of prioritization, the specific concepts, grouped into their respective categories (primary and secondary), are presented in order of their relevance (from most to least relevant) to project observatories, as illustrated in Figure 5.

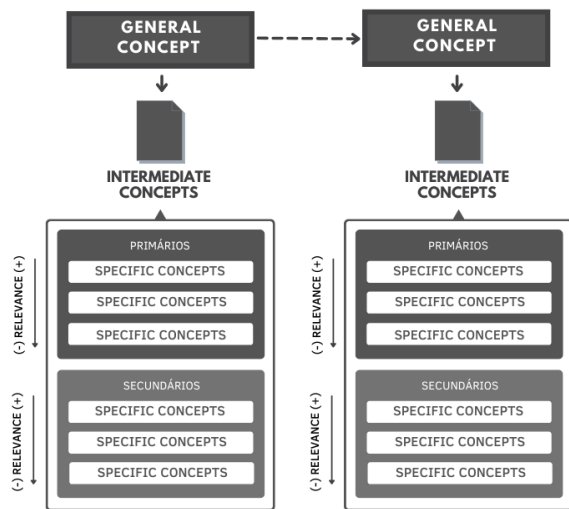


FIGURE 5. Model structure.

To provide a clearer understanding of the model, a fictitious scenario of a project observatory was developed. Using this scenario, an effort was made to identify how the observatory in question encompassed the concepts presented in the MPO.

2) OVERVIEW OF THE MPO

The fourth version of the MPO (presented in Figure 6) consists of 3 general concepts, 8 intermediate concepts, and 50 specific concepts. These concepts are organized in a vertical hierarchy with subordinating relationships. Additionally, Figure 6 shows the existence of horizontal relationships between the general concepts (represented by dashed arrows).

a: GENERAL CONCEPT "STRUCTURES"

This concept provides a structural and static view of a project observatory. The following intermediate concepts are grouped under Structures:

- Components: The constituent parts of a project observatory that support the processes executed within (or by) the observatory.
- Contents: Related to the subjects addressed by the project observatory. This concept helps identify the data, information, and knowledge that a project observatory collects, stores, processes, and/or shares.
- Characteristics: Pertains to a set of properties of a project observatory. These characteristics contribute to a better understanding of its processes, components, contents, and infrastructure.
- IT Infrastructure: The technological framework that provides the foundational support for a project observatory. This infrastructure comprises physical and virtual resources that support the flow, storage, processing, and analysis of data.

Figure 7 presents the intermediate and specific concepts grouped under "Structures." The description of each

specific concept included in "Structures" can be found in Appendix K.

b: GENERAL CONCEPT "PROCESSES"

This general concept presents a project observatory from a dynamic perspective, encompassing intermediate and specific concepts related to the processes that can be executed in or by a project observatory. These processes correspond to actions performed by or through the components of an observatory. The actors in an observatory may participate in these processes either by executing them or by receiving the results of these executions. Thus, these processes are grouped into:

- Managing Data and Information: this encompasses a set of processes related to the collection, storage, processing, and dissemination of data and information by a project observatory.
- Producing Knowledge and Intelligence: this includes a set of processes that, based on the data and information collected about the projects, are capable of aiding in the generation of knowledge and intelligence to meet the demands of the actors involved in a project observatory.

Figure 8 presents the intermediate and specific concepts grouped under "Processes." The description of each specific concept included in "Processes" can be found in Appendix K.

c: GENERAL CONCEPT "AGENTS"

This concept presents a project observatory from the perspective of its actors and the reasons that drive them to interact with the observatory. This general concept groups the following intermediate concepts:

- Actors: these correspond to the entities (human or system) that, in some way, maintain a relationship (direct or indirect) with a project observatory.
- Motivations: these refer to a set of reasons that lead the actors to maintain a relationship with a project observatory. A project observatory must have resources that meet these motivations.

Figure 9 presents the intermediate and specific concepts grouped under "Agents." The description of each specific concept included in "Agents" can be found in Appendix K.¹

VI. DISCUSSION OF RESULTS

The MPO conceptualizes project observatories as tools that enable a community to monitor and track projects of interest, thereby promoting transparency. Additionally, these observatories are also understood as tools that support the creation and sharing of knowledge about projects, which can be useful in decision-making processes within this context. Another significant characteristic of project observatories, as considered by the MPO, is the potential for

¹An alternative version of the model's figures is presented in the Appendix M, in the form of UML package diagrams.

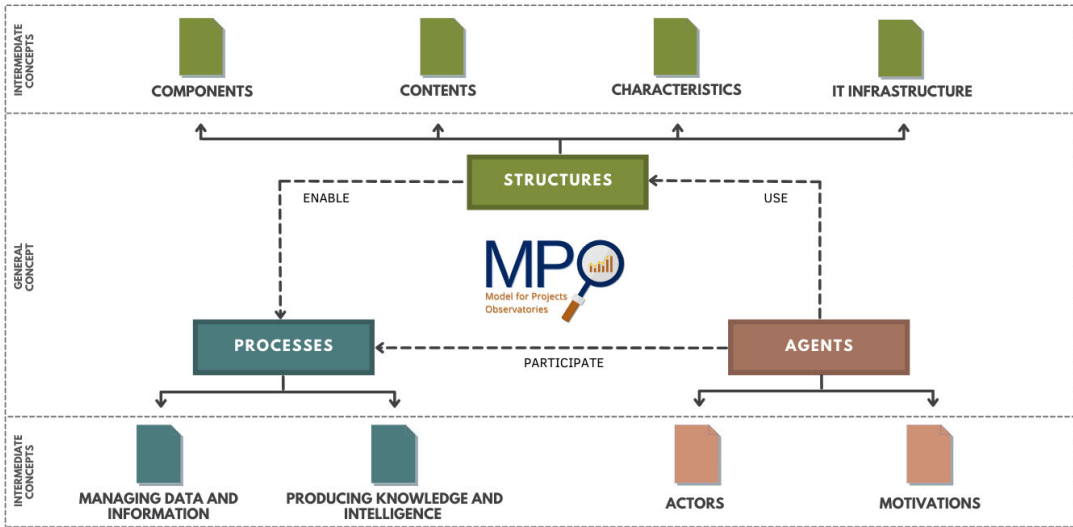


FIGURE 6. General and intermediate concepts.

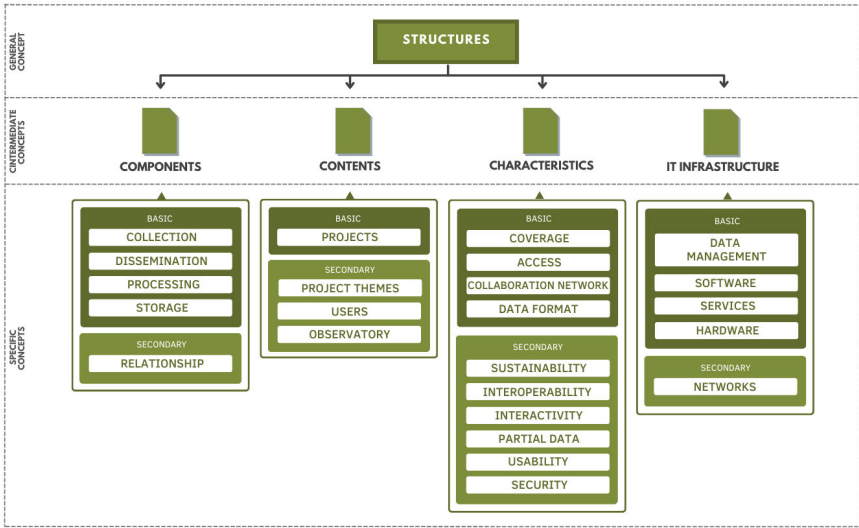


FIGURE 7. General concept “Structures.”

creating a collaborative network among stakeholders, promoting dialogue and interaction. Observation, transparency, knowledge creation and sharing, decision-making, and collaboration are relevant constructs in understanding observatories according to the available literature on the subject [26], [27].

The MPO was compared with other observatory models available in the literature [14], [15], [16], [55]. The most significant difference of the model proposed in this work, compared to other models, is the definition of “Projects” as the area of application for the observatories. This definition was crucial in identifying and detailing the concepts (especially the specific concepts) covered by the MPO.

Concerning structural elements, the MPO includes the most common elements found in other models, such as

structures, actors, and processes. However, a considerable difference is observed between the MPO and other models in terms of the detailing and decomposition of these elements into sub-elements. To confirm this assertion, a mapping between the models was conducted to identify the specific concepts proposed in the MPO that were covered (partially or fully) by other models (this mapping is discussed in detail in Appendix L).

The results of this mapping showed that none of the models fully covered all the specific concepts discussed in the MPO. These results reinforce the need (identified as the primary objective of this study) to develop a model for observatories in the context of projects, as none of the discussed models adequately covered this application context.

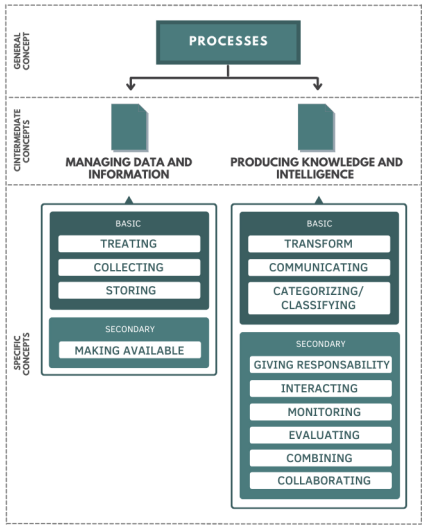


FIGURE 8. General concept “Processes.”

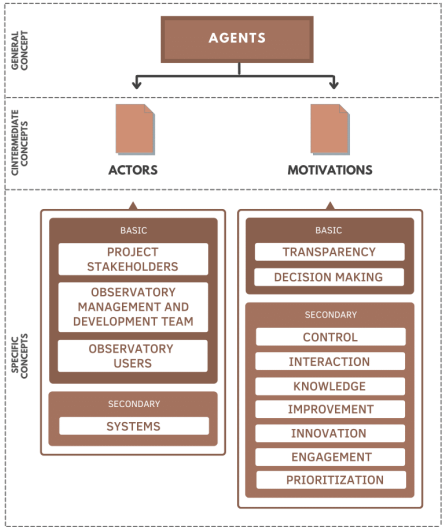


FIGURE 9. General concept “Agents.”

VII. CONCLUSION

The following research objective was defined to guide the development of this study: to propose a conceptual model that contributes to the understanding and development of project observatories, addressing aspects related to their structures, processes, and agents. To achieve this main objective, seven specific objectives were outlined.

The first specific objective aimed to understand how observatories are developed and characterized. To achieve this objective, a Systematic Literature Mapping (SLM) on observatories was conducted, investigating and understanding the state of the art of observatories by identifying their application areas, definitions, characteristics, objectives, benefits, challenges faced, components, and, finally, the products and services offered by these observatories.

The second, third, and fourth specific objectives were related to identifying the structural aspects, processes, and actors (and their motivations) of project observatories, respectively. To meet these objectives, ten pilot projects of project observatories were executed. These pilot projects allowed the identification of a set of facets related to structures, processes, and agents. These facets were confirmed by the systematic literature mapping, leading to the creation of the first version of the MPO, thus achieving the fifth specific objective proposed in this study: to design a conceptual model for project observatories.

The sixth specific objective of this study involved the evaluation and evolution of the proposed model. To meet this objective, methodological steps were defined, including three focus group sessions with project and observatory experts, a survey with participants from the pilot projects and observatory experts, and, finally, a multiple-case study where two project observatories were developed using the MPO as a reference. As a result of this evaluation process, three more versions of the model were produced.

The most significant contribution of this research is the proposition of the Model for Project Observatories (MPO), a conceptual model that can contribute both to the understanding and the development of project observatories. In the theoretical context, the MPO structures a body of knowledge about project observatories, pointing out concepts that, when analyzed together, can help form an understanding of these observatories. In the practical context, the results presented indicate that the MPO can serve as a reference and guidance source in the design processes of project observatories.

The limitations of this research include: (1) the Initial Exploratory Studies, where the involvement of postgraduate students might have affected the engagement and perceptions collected, suggesting the need to conduct studies in different contexts to gain new perspectives; (2) in the Systematic Literature Mapping, the limited number of backward and forward snowballing iterations due to the high return of references is a limitation, although the quantity and quality of the selected works mitigate this problem; (3) in the Focus Groups, the limited session time might have made it difficult to perform a detailed evaluation of the MPO artifact, especially with many experts participating, making the method costly for replication in multiple sessions; (4) in the Survey, developing a comprehensive questionnaire that was not overly lengthy was a challenge, as well as reaching a representative sample of the population; (5) in the Case Study, the fact that the projects were developed by students in academic settings and the potential bias introduced by the researcher acting as a participant observer are limitations to consider.

Finally, it is believed that this work can be expanded and, consequently, evolved through the development of new studies. These future directions include the development of new project observatories, the creation of a guide for developing project observatories, studies on interactivity in

the context of project observatories, and the development of computational solutions to support the development of project observatories.

APPENDIX A

LIST OF STUDIES INCLUDED IN THE SYSTEMATIC LITERATURE MAPPING

The content of this appendix can be accessed through the following link: <https://doi.org/10.5281/zenodo.15626686>

APPENDIX B

TRACEABILITY OF THE FIRST VERSION OF THE MPO WITH THE SLM

The content of this appendix can be accessed through the following link: <https://doi.org/10.5281/zenodo.15626686>

APPENDIX C

FACETED ANALYSIS OF INITIAL EXPLORATORY STUDIES

The content of this appendix can be accessed through the following link: <https://doi.org/10.5281/zenodo.15626686>

APPENDIX D

FIRST VERSION OF THE MPO

The content of this appendix can be accessed through the following link: <https://doi.org/10.5281/zenodo.15626686>

APPENDIX E

QUESTIONNAIRES USED IN THE FOCUS GROUP

The content of this appendix can be accessed through the following link: <https://doi.org/10.5281/zenodo.15626686>

APPENDIX F

SECOND VERSION OF THE MPO

The content of this appendix can be accessed through the following link: <https://doi.org/10.5281/zenodo.15626686>

APPENDIX G

QUESTIONNAIRE USED IN THE SURVEY

The content of this appendix can be accessed through the following link: <https://doi.org/10.5281/zenodo.15626686>

APPENDIX H

THIRD VERSION OF THE MPO

The content of this appendix can be accessed through the following link: <https://doi.org/10.5281/zenodo.15626686>

APPENDIX I

DETAILED RESPONSES TO THE CASE STUDY RESEARCH QUESTIONS

The content of this appendix can be accessed through the following link: <https://doi.org/10.5281/zenodo.15626686>

APPENDIX J

MPO CONCEPTS ADDRESSED BY THE CASE STUDY

The content of this appendix can be accessed through the following link: <https://doi.org/10.5281/zenodo.15626686>

APPENDIX K

FOURTH VERSION OF THE MPO

The content of this appendix can be accessed through the following link: <https://doi.org/10.5281/zenodo.15626686>

APPENDIX L

MAPPING BETWEEN THE MPO AND OTHER OBSERVATORY MODELS

The content of this appendix can be accessed through the following link: <https://doi.org/10.5281/zenodo.15626686>

APPENDIX M

MAPPING BETWEEN THE MPO AND OTHER OBSERVATORY MODELS

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APPENDIX N

ALTERNATIVE VERSION OF THE MODEL FIGURES USING UML PACKAGE DIAGRAMS

The content of this appendix can be accessed through the following link: <https://doi.org/10.5281/zenodo.15626686>

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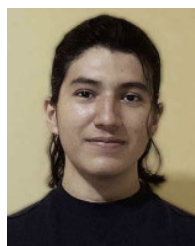
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